

AI-powered Denoising: Supervised Learning

The denoising problem

Estimate this



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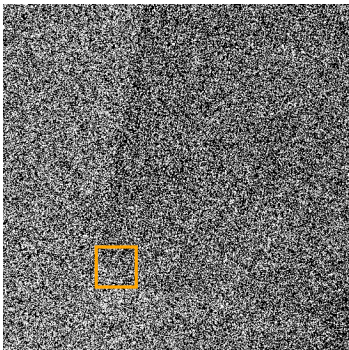


From this



Why?

Why?



How do we evaluate a denoiser?

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Mean squared error

$$\left\| \begin{array}{c} \text{denoised} \\ \text{image} \end{array} - \begin{array}{c} \text{clean} \\ \text{image} \end{array} \right\|_2^2$$

The diagram illustrates the Mean Squared Error (MSE) calculation for evaluating a denoiser. It shows two grayscale images side-by-side, separated by a minus sign. The left image is labeled 'denoised' and the right image is labeled 'clean'. Above the images, the text 'MSE' is written, and to the right of the minus sign, a large '2' indicates the squared norm. The entire expression is enclosed in large vertical bars, representing the L2 norm.

How do we evaluate a denoiser?

Mean squared error

$$\left\| \begin{array}{c} \text{denoised} \\ \text{clean} \end{array} \right\|_{\text{MSE}}^2$$

The diagram illustrates the Mean Squared Error (MSE) calculation. It features two grayscale photographs of a man in a dark coat operating a vintage camera on a tripod. The left photograph is labeled 'denoised' and shows significant digital noise. The right photograph is labeled 'clean' and is clear. A horizontal minus sign is positioned between the two images. Above the images, the text 'MSE' is centered, and to the right of the images, a large superscript '2' indicates the squared magnitude of the difference.

Peak signal-to-noise ratio

$$\text{PSNR} := 10 \log \left(\frac{M^2}{\text{MSE}} \right)$$

M : peak signal value

How do we evaluate a denoiser?

Mean squared error

$$\left\| \begin{array}{c} \text{MSE} \\ \text{denoised} \end{array} - \begin{array}{c} \text{clean} \end{array} \right\|^2$$

Peak signal-to-noise ratio

$$\text{PSNR} := 10 \log \left(\frac{M^2}{\text{MSE}} \right)$$

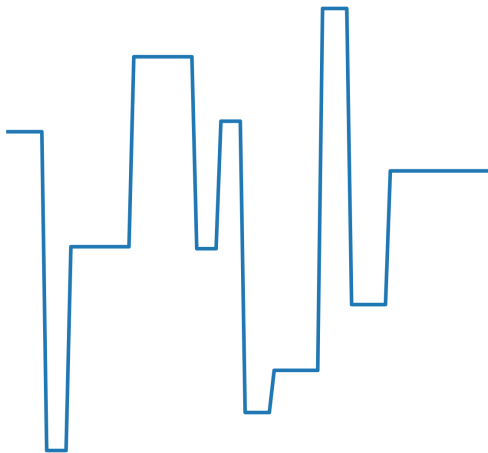
M : peak signal value

Warning: May not reflect application-specific performance

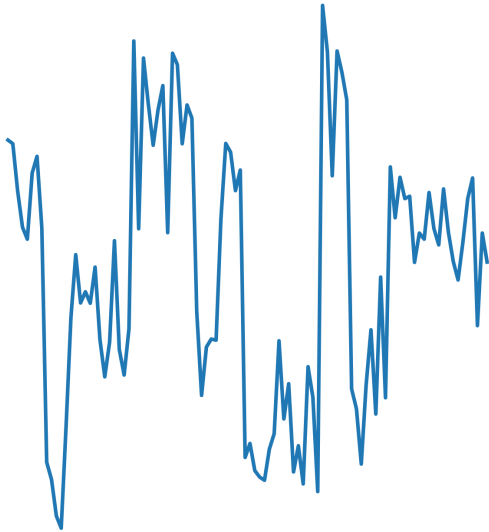
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

1D piecewise-constant functions



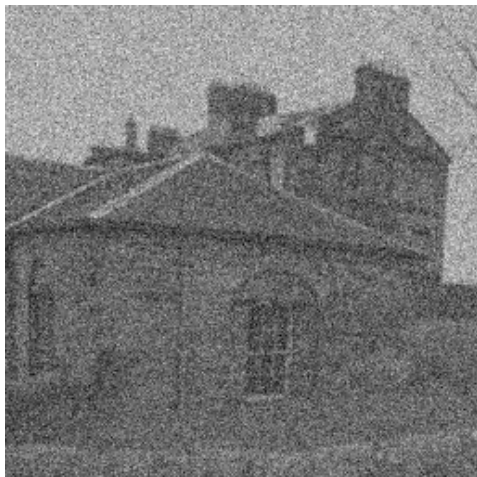
Additive i.i.d. Gaussian noise (PSNR 19.96 dB)



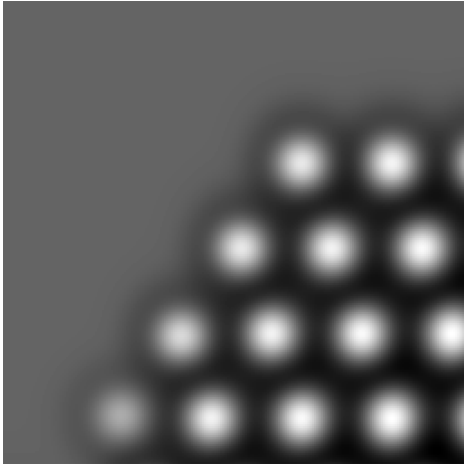
Natural images



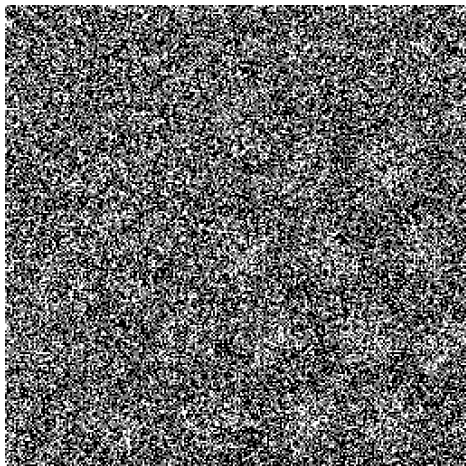
Additive i.i.d. Gaussian noise with variable variance
(PSNR 19.99 dB)



Electron microscopy



Poisson shot noise (PSNR 6.64 dB)



Plan

Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

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Challenge

How to estimate thousands of clean pixels from thousands of noisy pixels?

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Idea: Assume (1) linearity and (2) shift invariance

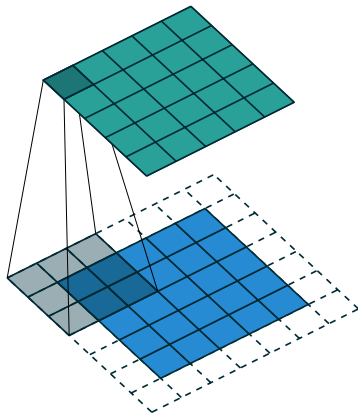
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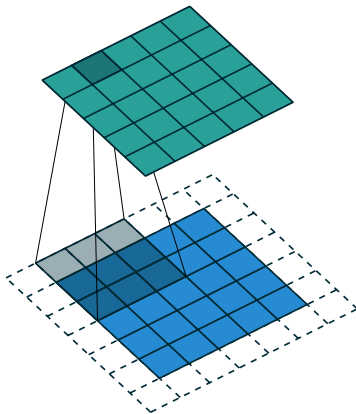
Optimal (Wiener) denoiser is convolutional

Convolutional filter



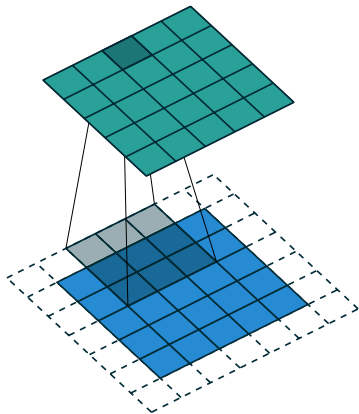
A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolutional filter



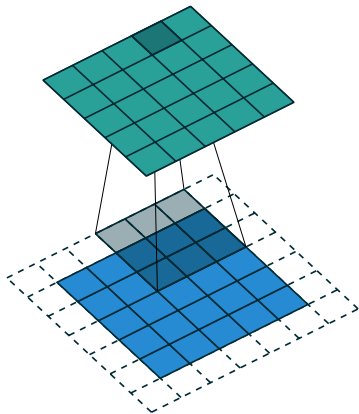
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Convolutional filter



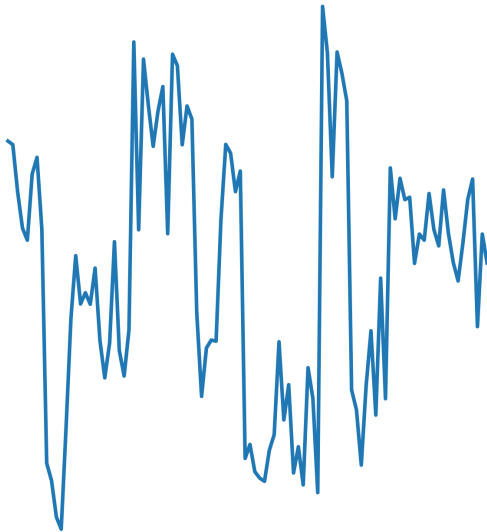
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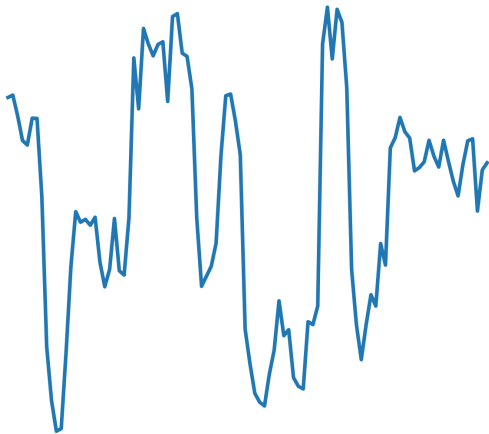


A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

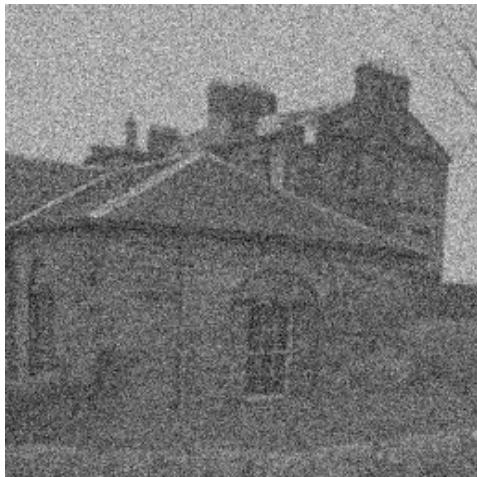
Noisy data (PSNR 19.96 dB)



LSI denoising (PSNR 22.33 dB)



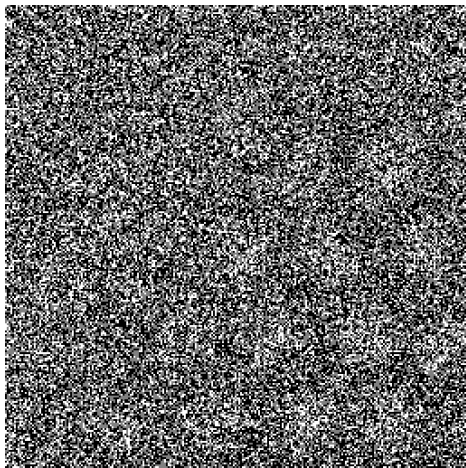
Noisy data (PSNR 19.99 dB)



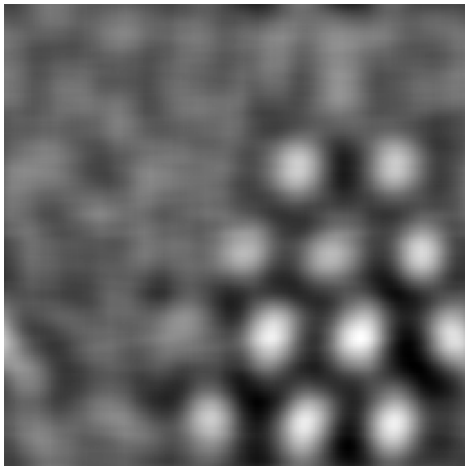
LSI denoising (PSNR 25.78 dB)



Noisy data (PSNR 6.64 dB)



LSI denoising (PSNR 29.80 dB)



Limitations of LSI denoising?

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Same filter applied at every location

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Same filter applied at every location

Blurs edges and other features

Pre-deep-learning solutions

Pre-deep-learning solutions

- ▶ Adapt filter locally (e.g. bilateral filter)

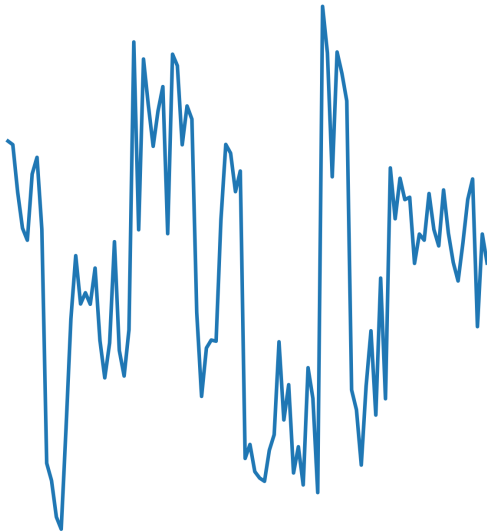
Pre-deep-learning solutions

- ▶ Adapt filter locally (e.g. bilateral filter)
- ▶ Exploit approximate sparsity of images in transformed domains (finite differences, wavelets...)

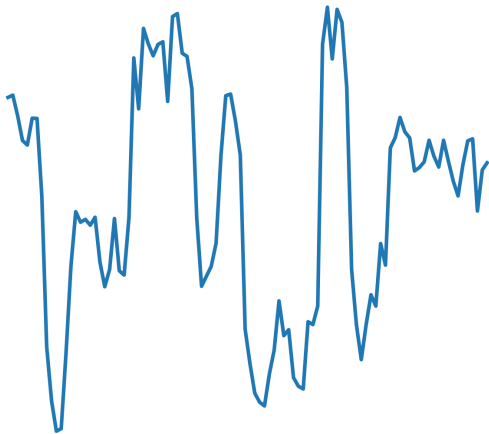
Pre-deep-learning solutions

- ▶ Adapt filter locally (e.g. bilateral filter)
- ▶ Exploit approximate sparsity of images in transformed domains (finite differences, wavelets...)
- ▶ Exploit similarity between image patches to perform nonlocal averaging or collaborative filtering

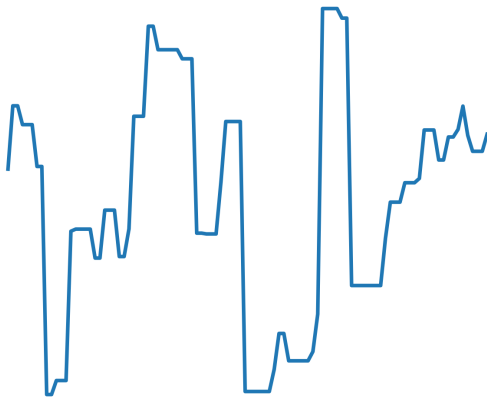
Noisy data (PSNR 19.96 dB)



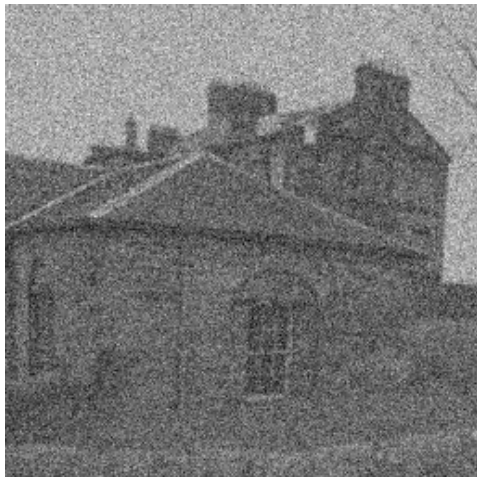
LSI denoising (PSNR 22.33 dB)



Total variation denoising (PSNR 25.07 dB)



Noisy data (PSNR 19.99)



LSI denoising (PSNR 25.78)

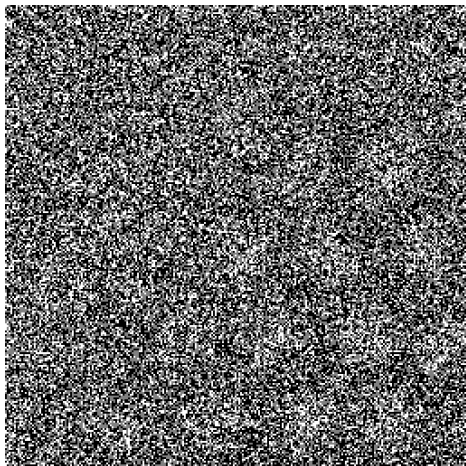


BM3D denoising (PSNR 30.62)

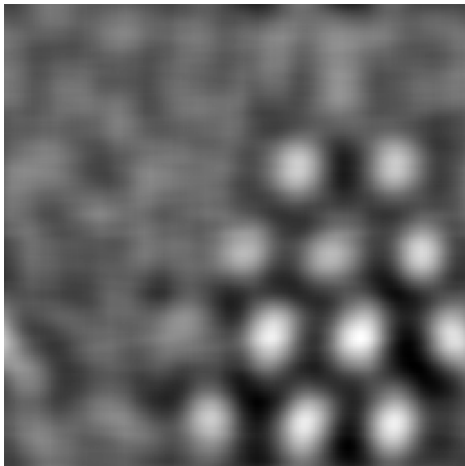


Image denoising by sparse 3-D transform-domain collaborative filtering.
Dabov et al. 2007

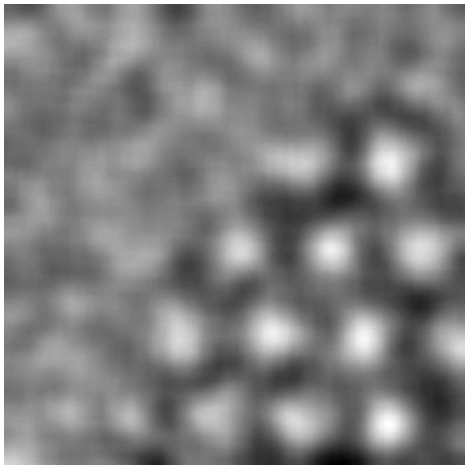
Noisy data (PSNR 6.64)



LSI denoising (PSNR 29.80)



PURE-LET denoising (PSNR 30.72)



A new SURE approach to image denoising: Interscale orthonormal wavelet thresholding. Luisier et al. 2007

Denoising before deep learning

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Conclusions (and a word of caution!)

Convolutional neural networks

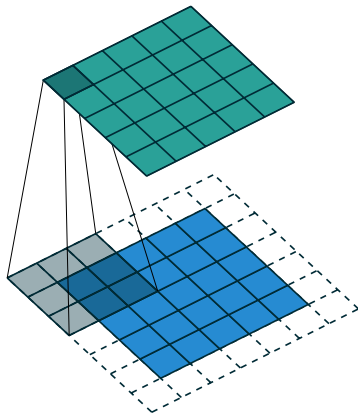
Designed to learn complex nonlinear functions from data

Convolutional neural networks

Designed to learn complex nonlinear functions from data

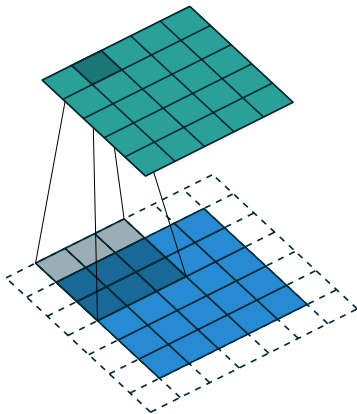
Convolutional layers interleaved with nonlinearities

Convolution



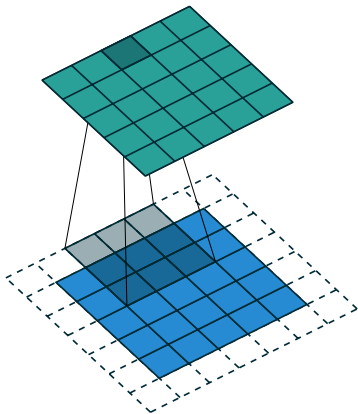
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Convolution



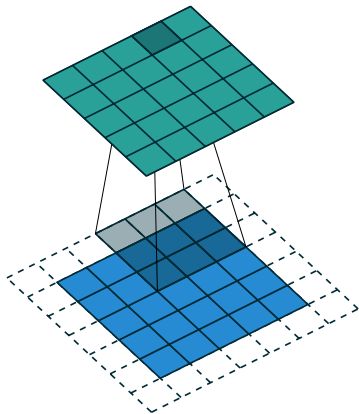
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Convolution



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ReLU

Rectifying linear unit

ReLU

Rectifying linear unit

Sets **negative** entries to **zero**

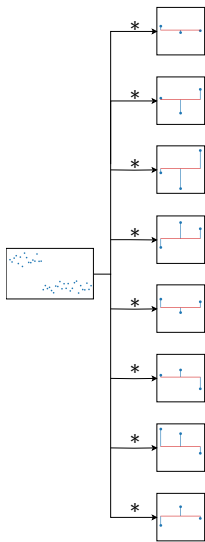
Convolutional neural network

Input



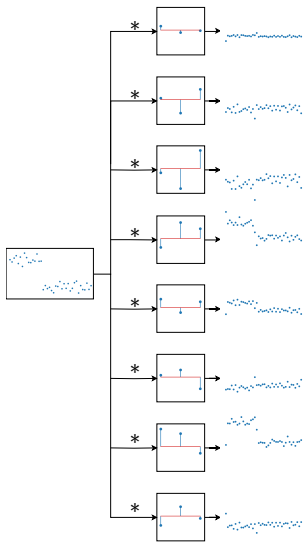
Convolutional neural network

Input Convolutional layer



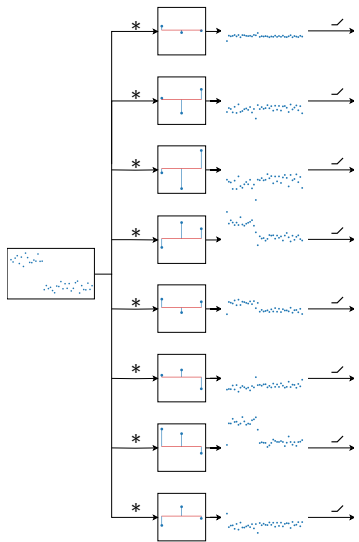
Convolutional neural network

Input Convolutional layer



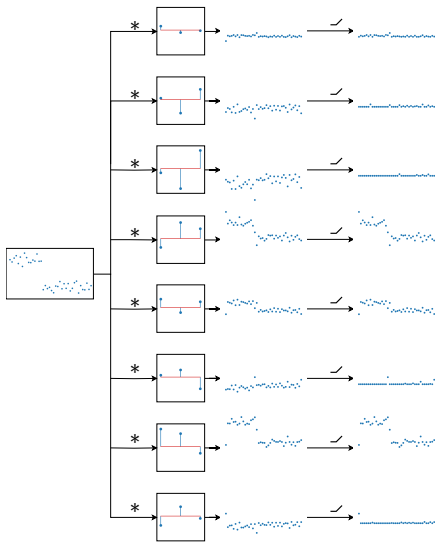
Convolutional neural network

Input Convolutional layer ReLU



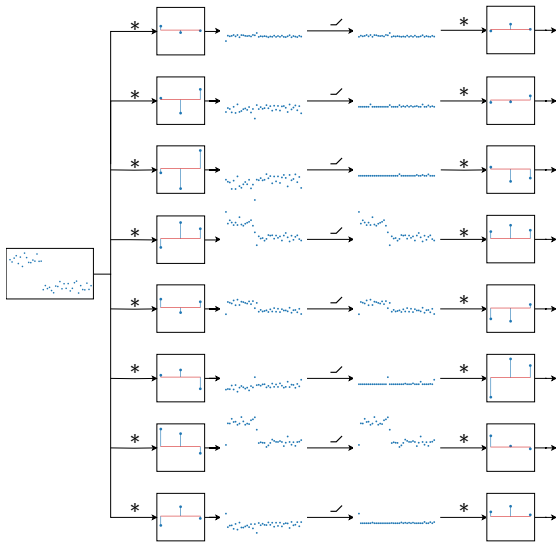
Convolutional neural network

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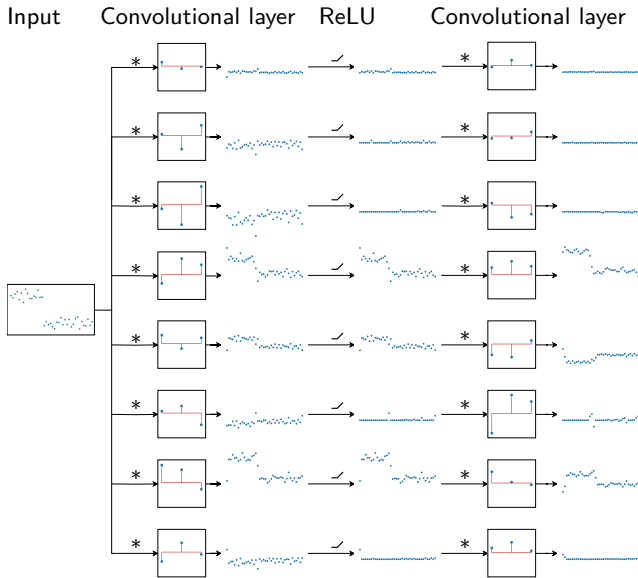


Convolutional neural network

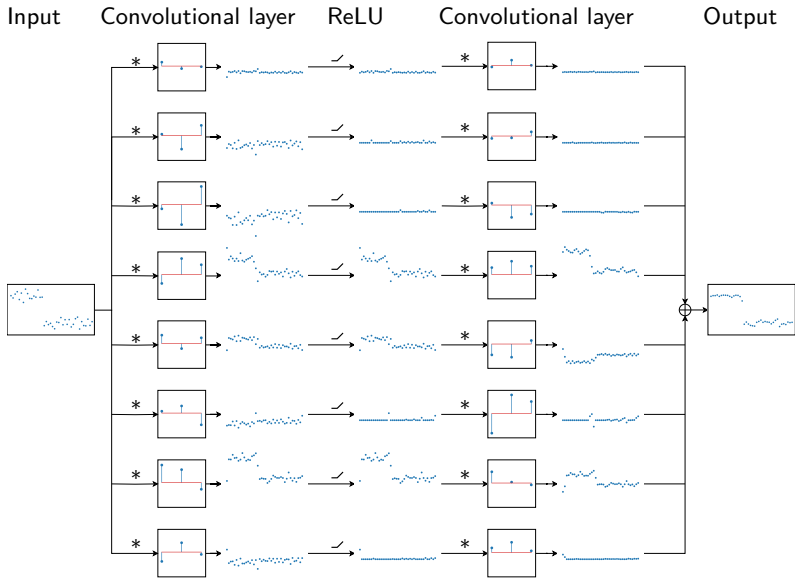
Input Convolutional layer ReLU Convolutional layer



Convolutional neural network



Convolutional neural network



Number of layers?

Number of layers?

Too few layers can result in underfitting

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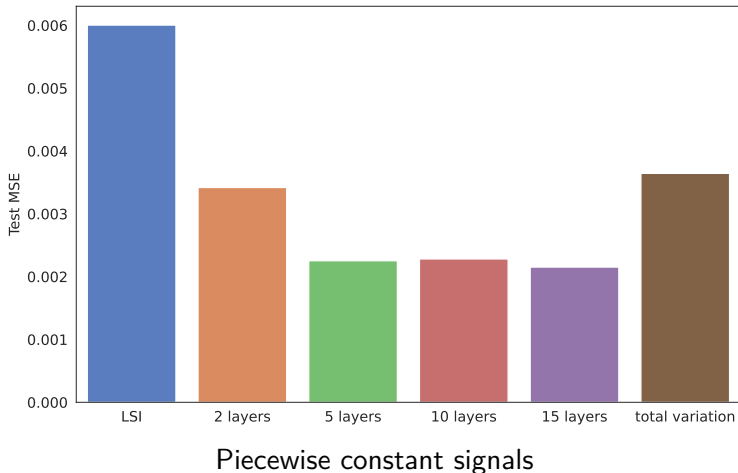
Too few layers can result in underfitting

Once the network is complex enough, diminishing returns

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Too few layers can result in underfitting

Once the network is complex enough, diminishing returns



Batch normalization

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Features in each channel are

1. Standardized, removing mean and dividing by standard deviation

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1. Standardized, removing mean and dividing by standard deviation
2. Re-centered and re-scaled with learned parameters

Skip connections

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Links between layers at different depths, bypassing the layers in between

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In denoising, a skip connection between input and output is often helpful

Skip connections

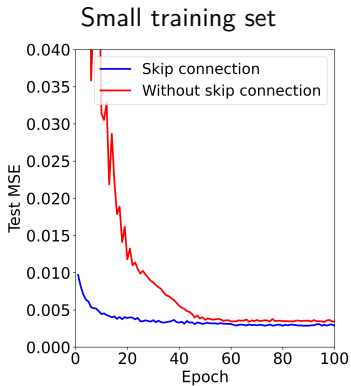
Links between layers at different depths, bypassing the layers in between

In denoising, a skip connection between input and output is often helpful

Equivalently, we learn a **residual** that is added to the noisy input

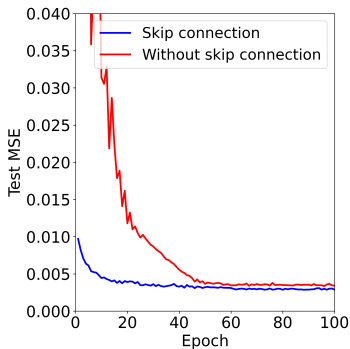
Piecewise constant signals

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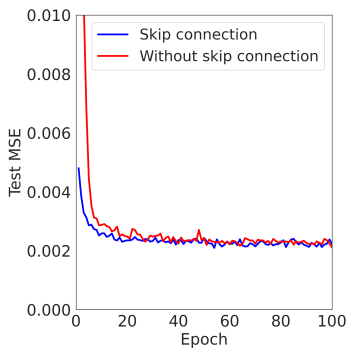


Piecewise constant signals

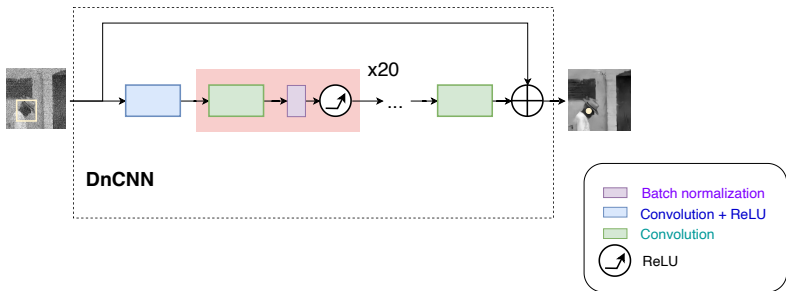
Small training set



Large training set

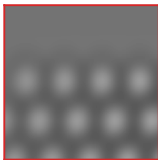
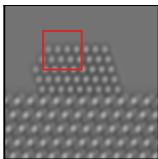


Denoising CNN (DnCNN)

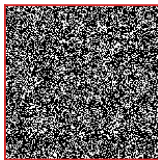
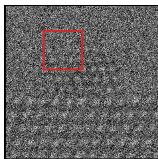


Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising. Zhang et al. 2017

Electron microscopy

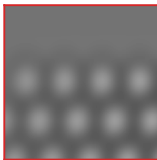
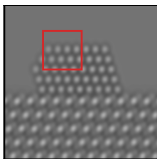


Simulated
clean image

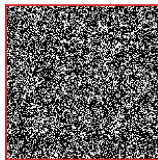
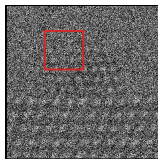


Noisy image

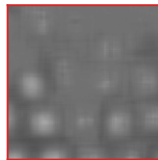
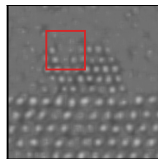
Electron microscopy



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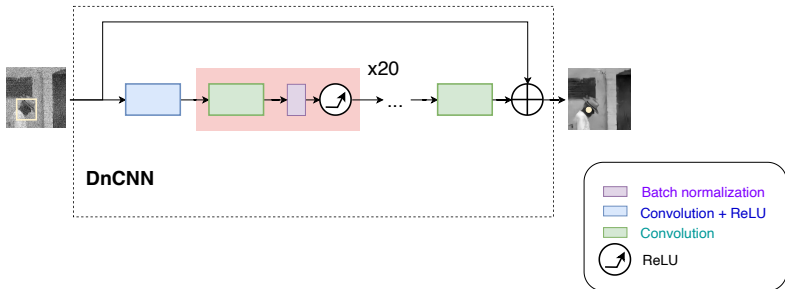


Noisy image

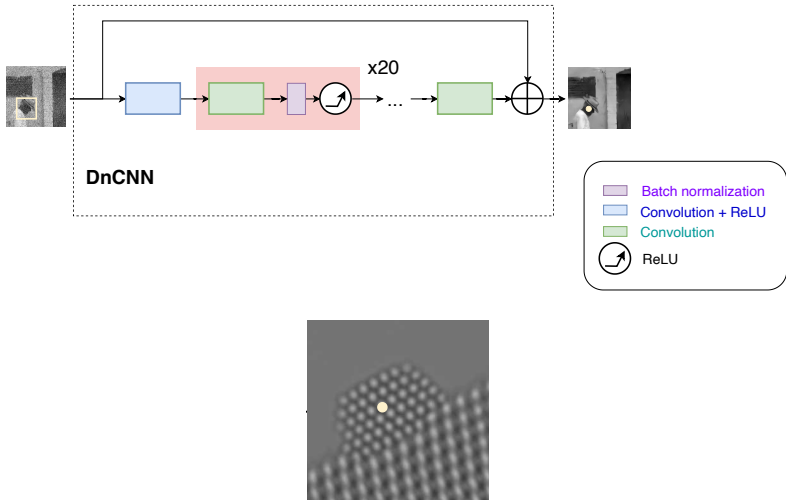


DnCNN

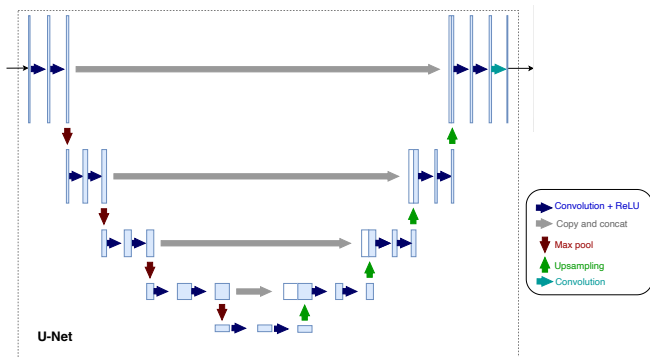
Receptive field



Receptive field



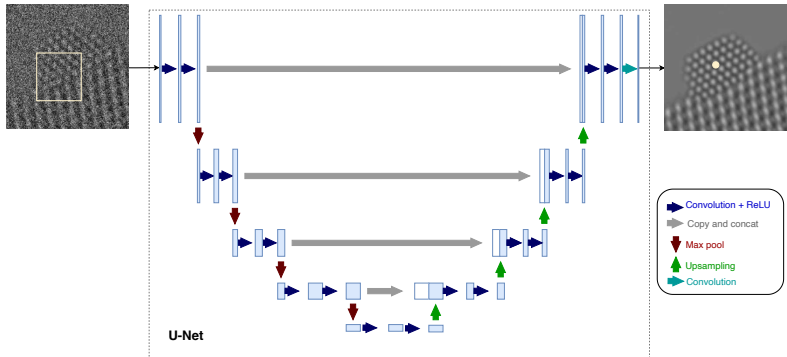
U-Net



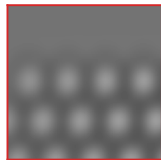
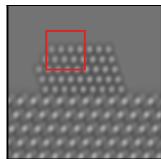
U-Net: convolutional networks for biomedical image segmentation.

Ronneberger et al. 2015

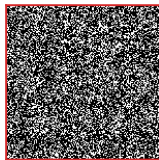
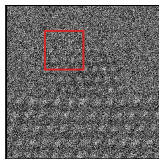
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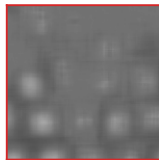
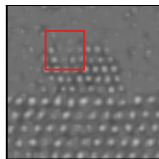
Electron microscopy



Simulated
clean image

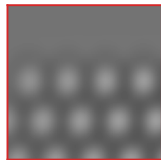
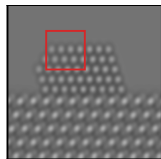


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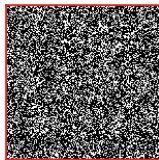
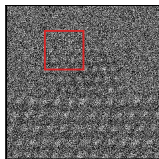


DnCNN

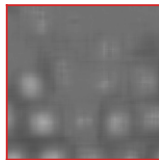
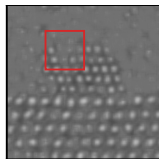
Electron microscopy



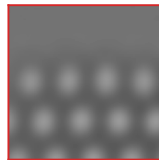
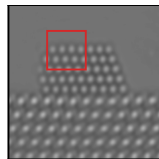
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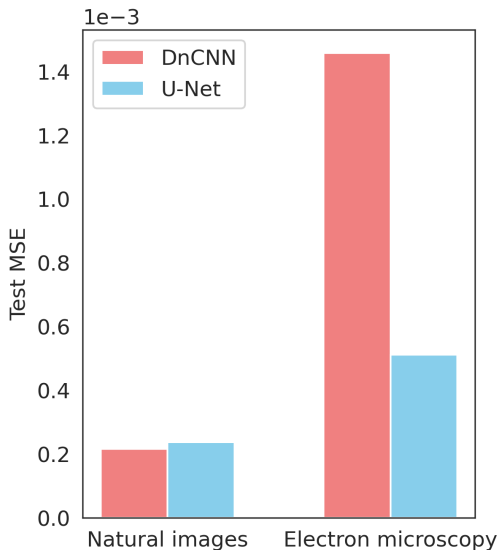


DnCNN



UNet

DnCNN vs U-Net



Denoising before deep learning

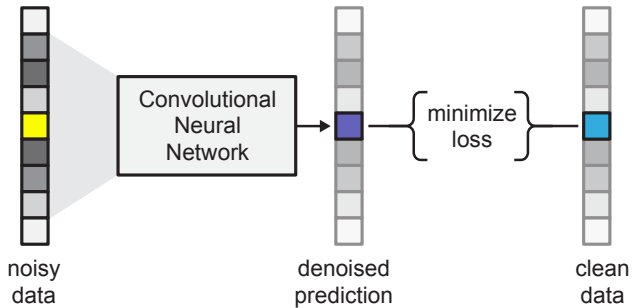
Convolutional neural networks

Supervised denoising

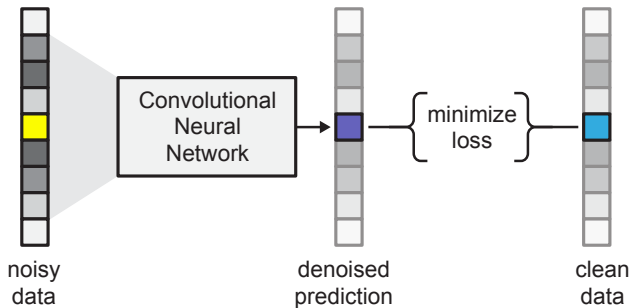
What do deep denoisers learn?

Conclusions (and a word of caution!)

Supervised learning

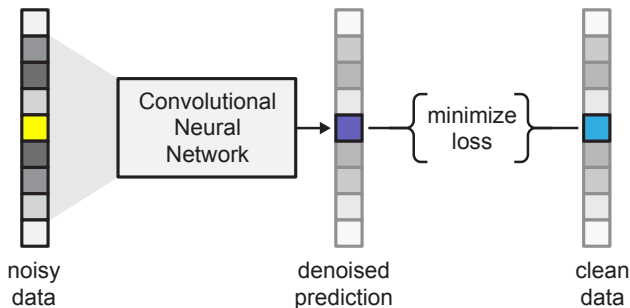


Supervised learning



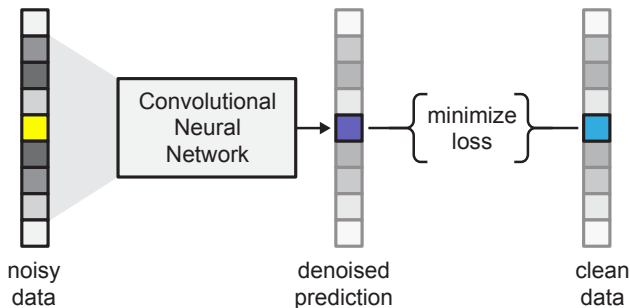
- Assumes access to clean - noisy pairs

Supervised learning



- ▶ Assumes access to clean - noisy pairs
- ▶ Minimizes mean squared error over training pairs via stochastic gradient descent

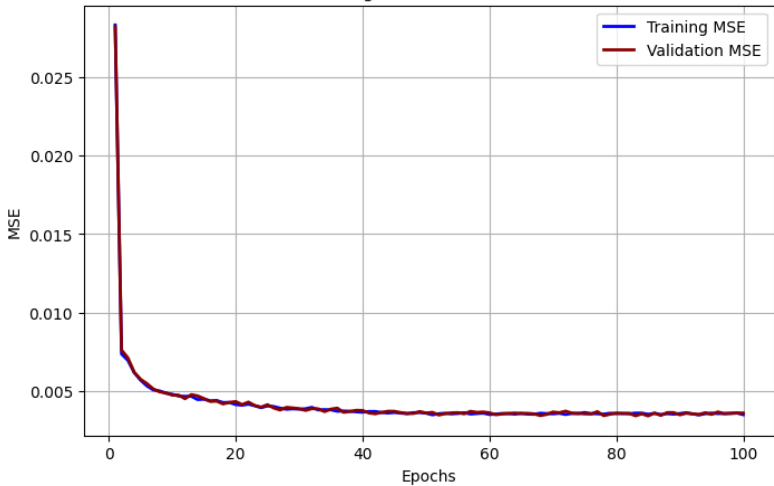
Supervised learning



- ▶ Assumes access to clean - noisy pairs
- ▶ Minimizes mean squared error over training pairs via stochastic gradient descent
- ▶ Gradient updates computed via backpropagation

Piecewise constant functions

Training and Validation MSE



Continuous noise sampling

If we know the noise distribution, we can simulate it
continuously during training

Continuous noise sampling

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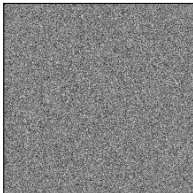


Continuous noise sampling

If we know the noise distribution, we can simulate it
continuously during training



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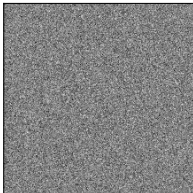


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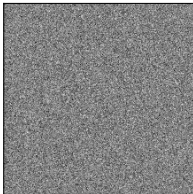


Continuous noise sampling

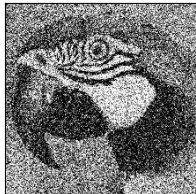
If we know the noise distribution, we can simulate it **continuously** during training



+

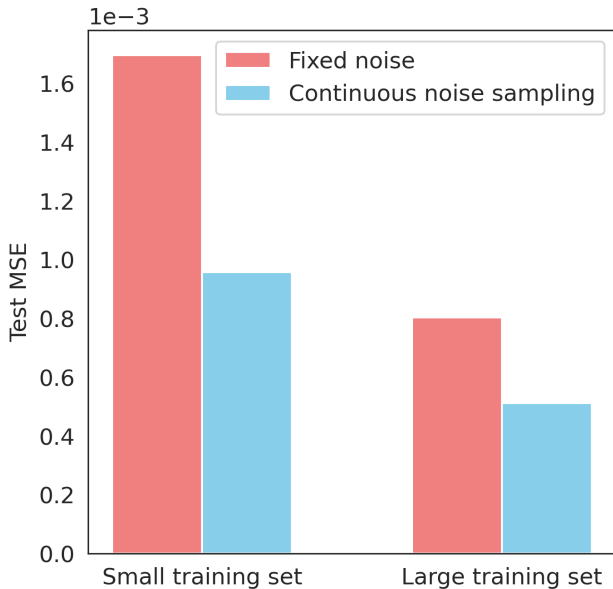


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Prevents **overfitting** specific noise samples

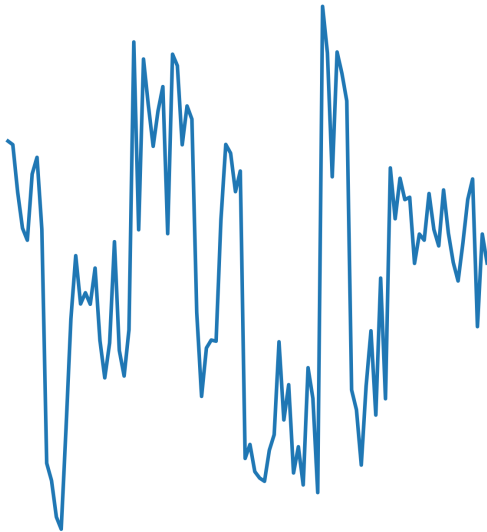
Electron microscopy data



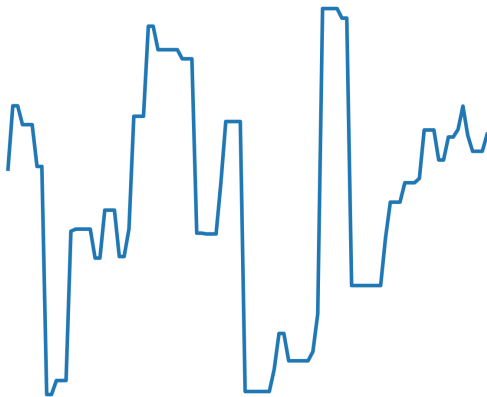
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

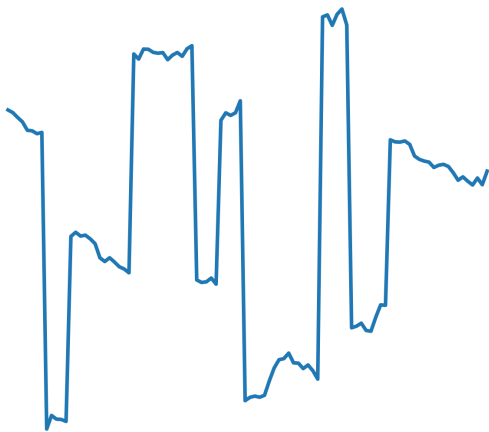
Noisy data (PSNR 19.96 dB)



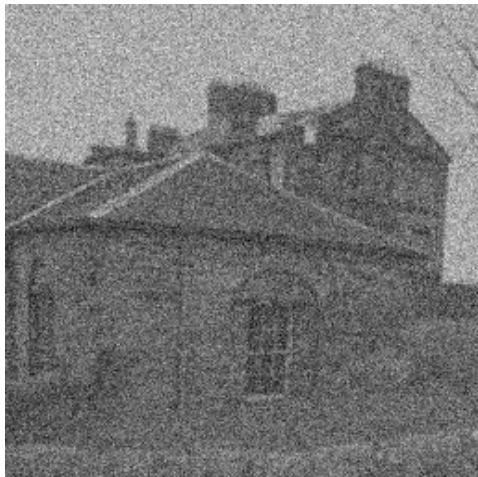
Total variation denoising (PSNR 25.07 dB)



Neural network (PSNR 26.44 dB)



Noisy data (PSNR 19.99 dB)



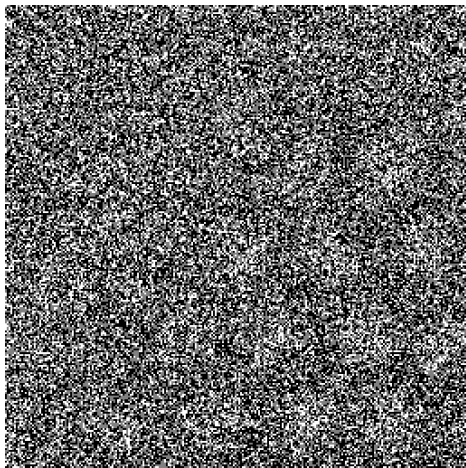
BM3D denoising (PSNR 30.62 dB)



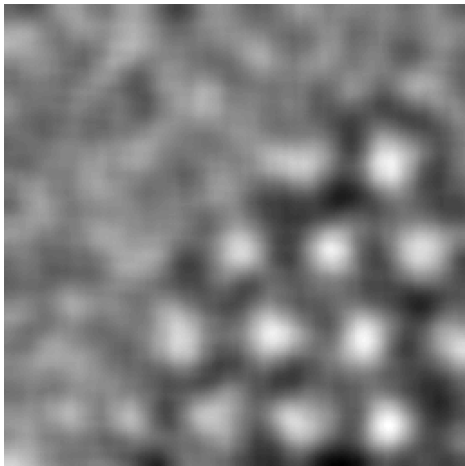
Neural network (PSNR 30.78 dB)



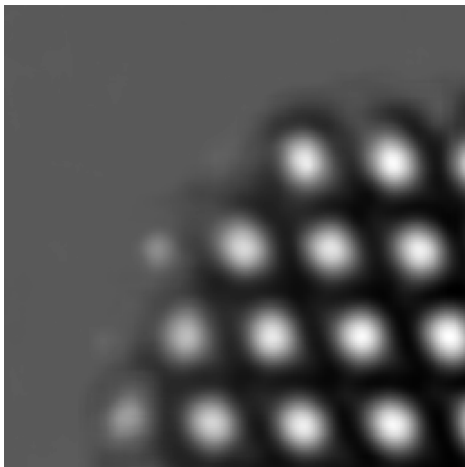
Noisy data (PSNR 6.64 dB)



PURE-LET denoising (PSNR 30.72 dB)



Neural network (PSNR 34.78 dB)



Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

How do neural networks denoise?

They implement complex nonlinear functions that depend on the input

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Gradient of output with respect to input reveals learned denoising mechanism *for that specific input*

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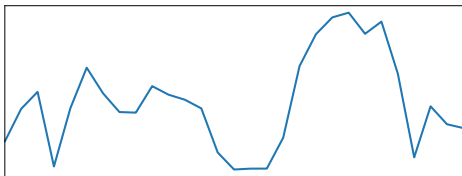
Gradient of output with respect to input reveals learned denoising mechanism *for that specific input*

Robust and interpretable blind image denoising via bias-free convolutional neural networks. Mohan et al. 2020

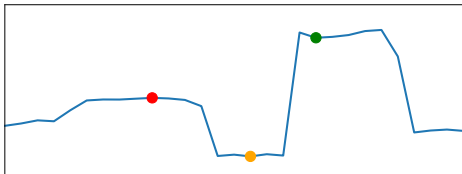
The line graph displays the daily death toll from COVID-19 in the United States. The x-axis represents time from March 2020 to March 2021, with labels for March, May, July, September, and November for each year. The y-axis represents the number of deaths, ranging from 0 to 120,000 in increments of 20,000. The graph shows a period of low mortality in early 2020, followed by a sharp increase starting in May 2020, peaking at approximately 105,000 deaths in late May/early June 2020. This is followed by a decline and then a second, even higher peak in early 2021, reaching over 100,000 deaths per day. After this peak, there is a sharp decline in late 2020 and a subsequent rise in early 2021.

The graph displays a blue line with three specific points highlighted. A red dot is located at a local maximum on the left side of the plot. A yellow dot is located at a local minimum in the center of the plot. A green dot is located at a sharp peak on the right side of the plot.

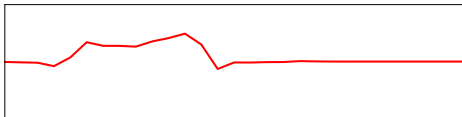
Noisy data



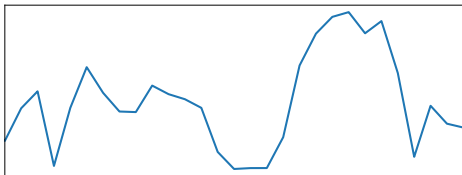
Denoised estimate



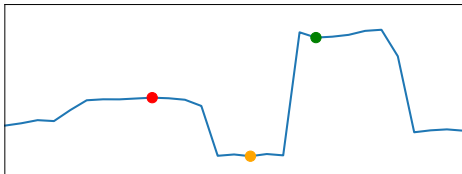
Gradient with respect to input



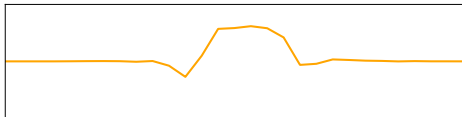
Noisy data y



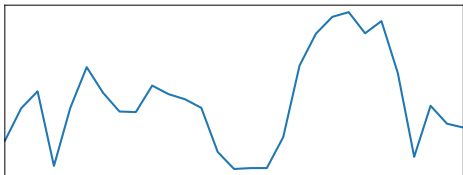
Denoised estimate $\mathcal{D}_\Theta(y)$



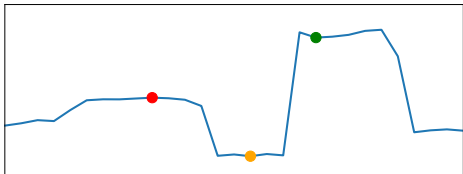
Gradient with respect to input



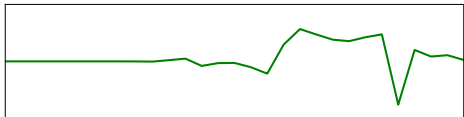
Noisy data y



Denoised estimate $\mathcal{D}_{\Theta}(y)$

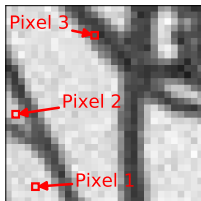


Gradient with respect to input

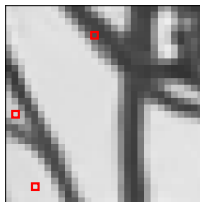


Low noise

Noisy image

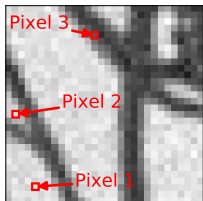


Denoised

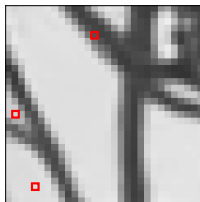


Low noise

Noisy image



Denoised

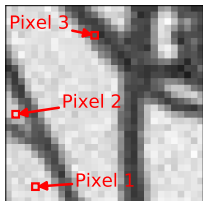


Pixel 1

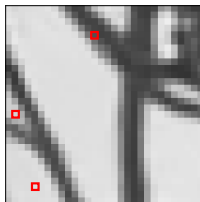


Low noise

Noisy image



Denoised



Pixel 1

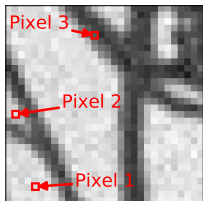


Pixel 2

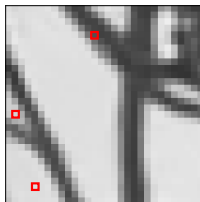


Low noise

Noisy image



Denoised



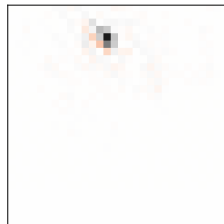
Pixel 1



Pixel 2

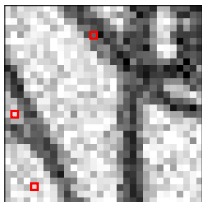


Pixel 3

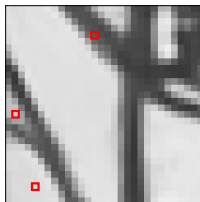


Medium noise

Noisy image

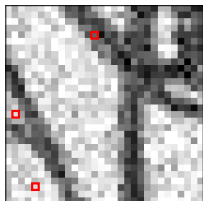


Denoised

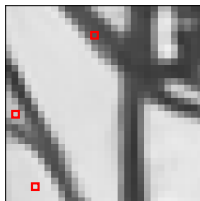


Medium noise

Noisy image



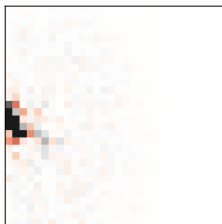
Denoised



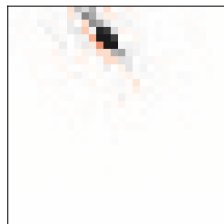
Pixel 1



Pixel 2

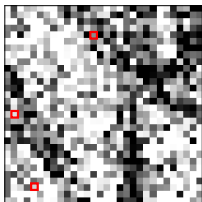


Pixel 3

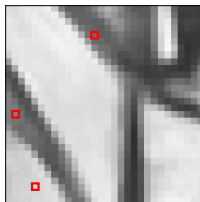


High noise

Noisy image

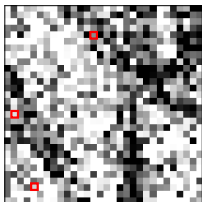


Denoised

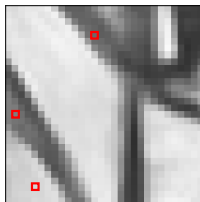


High noise

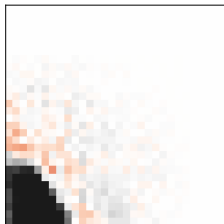
Noisy image



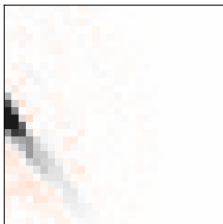
Denoised



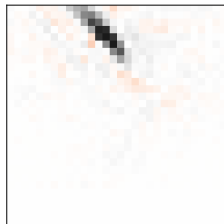
Pixel 1



Pixel 2

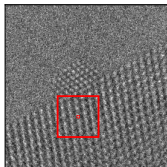


Pixel 3

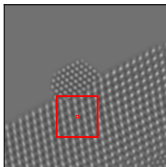


Electron microscopy

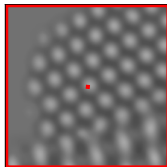
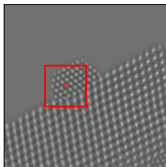
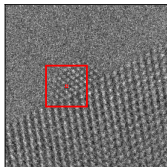
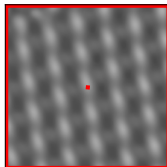
Noisy



Denoised

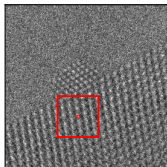


Denoised
(zoomed)

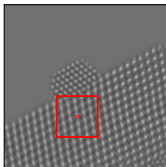


Electron microscopy

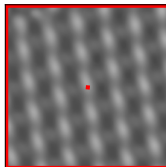
Noisy



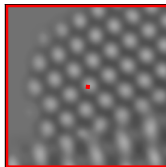
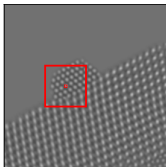
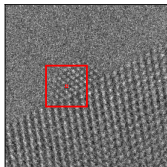
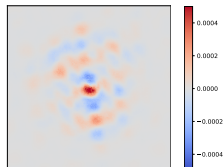
Denoised



Denoised
(zoomed)

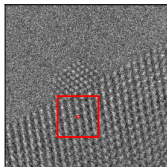


Gradient

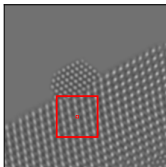


Electron microscopy

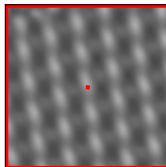
Noisy



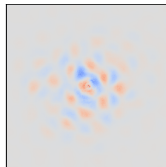
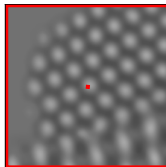
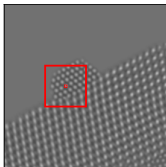
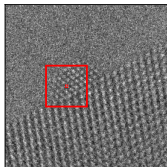
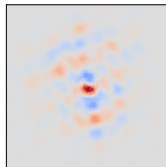
Denoised



Denoised
(zoomed)



Gradient



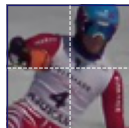
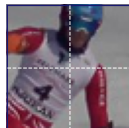
Video denoising



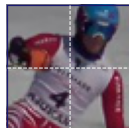
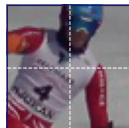
Video denoising



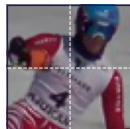
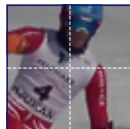
Video denoising



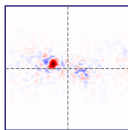
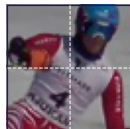
Video denoising



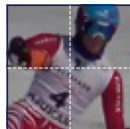
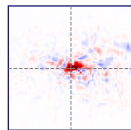
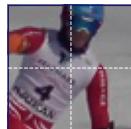
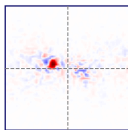
Video denoising



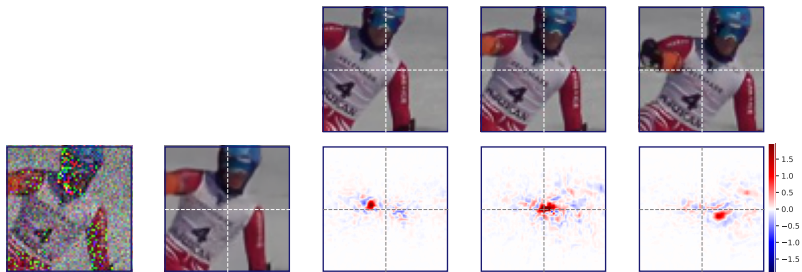
Video denoising



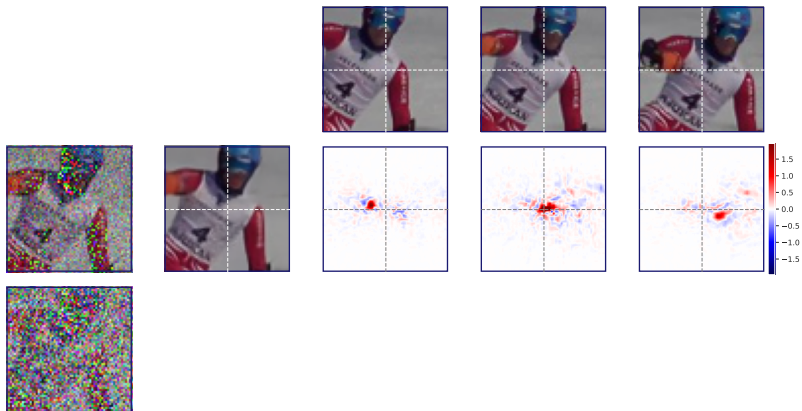
Video denoising



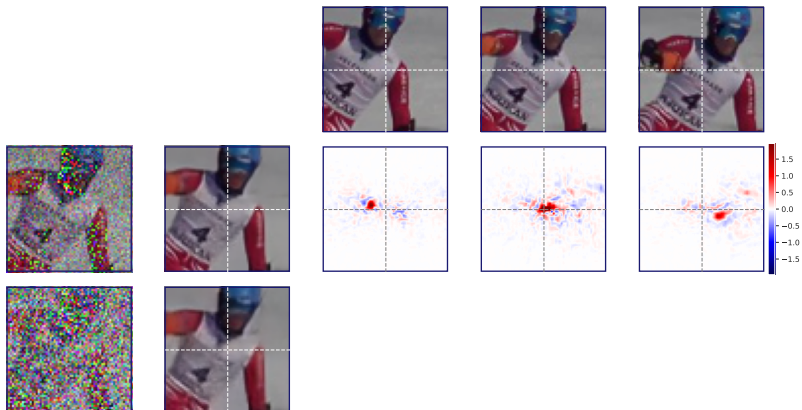
Video denoising



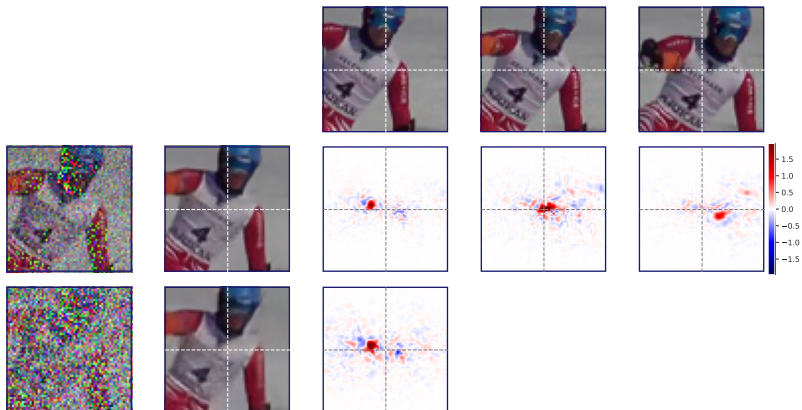
Video denoising



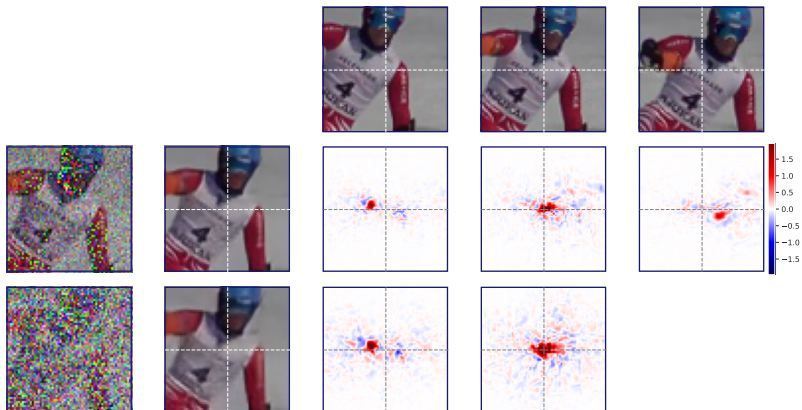
Video denoising



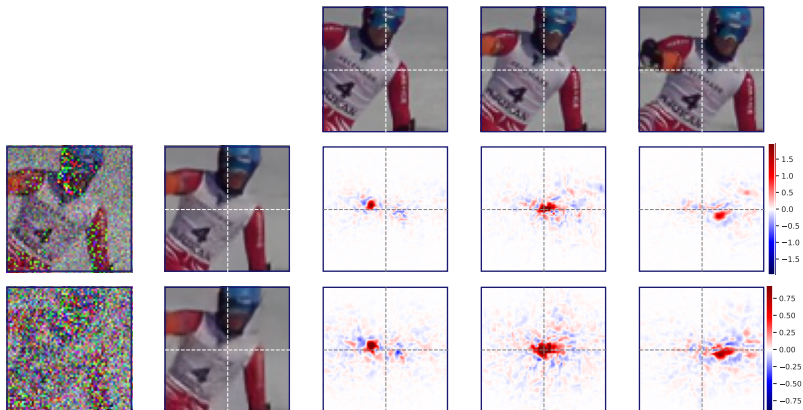
Video denoising



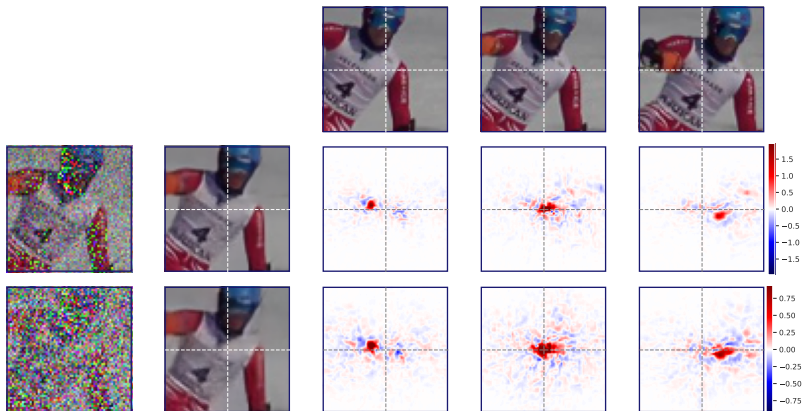
Video denoising



Video denoising



Video denoising



Unsupervised Deep Video Denoising. Sheth et al. 2021

Implicit motion compensation

(a) Noisy frame
($\sigma = 30$)



Implicit motion compensation

(a) Noisy frame
($\sigma = 30$)



(b) Motion estimate
from clean video



Implicit motion compensation

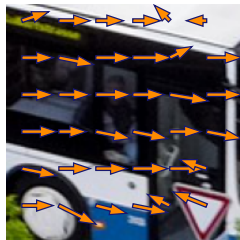
(a) Noisy frame
($\sigma = 30$)



(b) Motion estimate
from clean video



(c) Motion estimate
from network gradients



Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

Conclusions

Conclusions

Supervised denoising is *extremely effective*

Conclusions

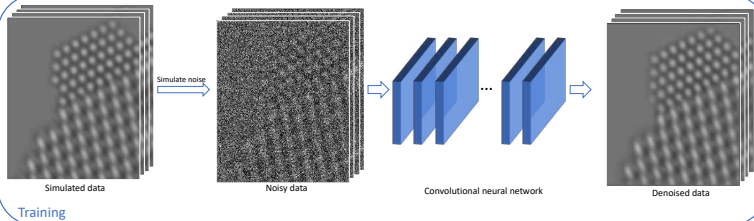
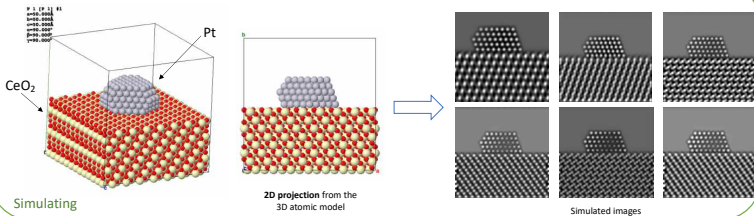
Supervised denoising is *extremely effective*

Models learn denoising strategies adapted to different signals and noise distributions **automatically**

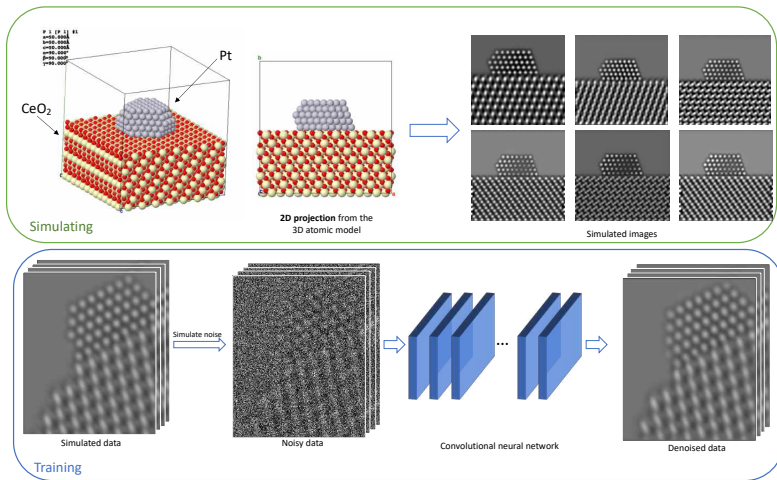
A word of caution

Distribution shift with respect to training data can cause models to fail!

Electron microscopy



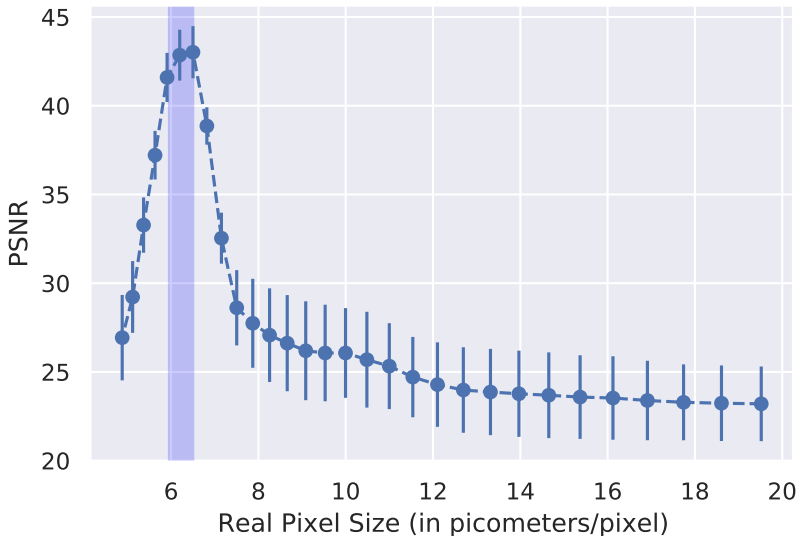
Electron microscopy



Experiment: Apply model to data at a different (1) resolution or (2) orientation

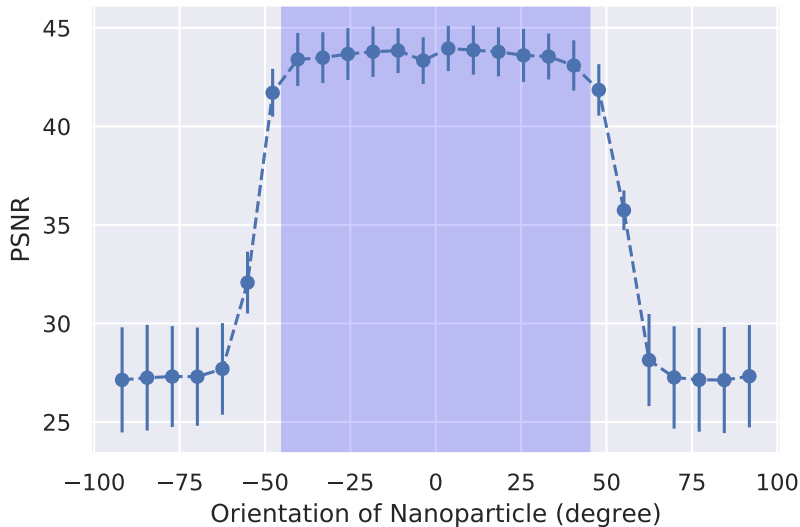
Resolution

Resolution



Orientation

Orientation



For more information

**Learn, Denoise and Discover: A Guide to Deep Denoising
with an Application to Electron Microscopy**

Sreyas Mohan, Kangning Liu, Peter Crozier,
Carlos Fernandez-Granda

Acknowledgements

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